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An exploration of colour and structure as used in the interface design of dominant video sharing platforms such as YouTube

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Abstract

This dissertation explores how colour and structure within user interfaces shape user perception, engagement, and behaviour on dominant video-sharing platforms, focusing particularly on YouTube. Through a qualitative and interpretive methodology combining semiotic theory, design principles, and psychological studies, this research analyses the emotional and cognitive roles of colour and structure in digital environments. A historical overview of colour symbolism establishes a cultural context, followed by an examination of branding strategies, interface hierarchy, and platform-specific visual design choices. Comparative analysis of YouTube, TikTok, Vimeo, and Instagram reveals how colour and navigation structures create distinct emotional rhythms and user experiences. Special attention is given to teenage users, highlighting how emotional and cognitive development stages amplify their sensitivity to colour dynamics. The study concludes that colour and structure function not merely as aesthetic features but as strategic semiotic tools that influence user interaction and retention, raising broader ethical considerations about emotional engineering in digital design.

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Introduction

In an increasingly digitalized world, the design of user interfaces has emerged as a critical factor in shaping user engagement, perception, and behaviour. Design functions as a purposeful act of communication as well as an aesthetic decision, especially when it comes to colour and structural organisation. Donald Norman aptly frames this perspective by stating that "design is really an act of communication, which means having a deep understanding of the person with whom the designer is communicating" (Norman, 2013). The combination of colour and structural components creates a complex semiotic system that both reflects and shapes user engagement in the context of popular video-sharing websites like YouTube. The aim of this dissertation is to explore how colour and structure are utilised within the interface design of major video-sharing platforms, with a specific focus on YouTube. In this situation, colour serves as a powerful semiotic instrument that communicates identification, emotion, and navigational clues. Content arrangement, page layout, and general navigational flow are all examples of structure, which has a direct effect on how users perceive information and feel. When combined, these components not only provide visual appeal but also strategically direct user behaviour, supporting platform objectives like higher engagement and retention.

The research objectives are as follows:

- To analyse how colour functions as a semiotic and emotional tool in video-sharing platform interfaces.
- To examine the role of structural design in influencing user behaviour, engagement, and retention.
- To compare the visual strategies of different platforms such as YouTube, TikTok, and Vimeo.
- To assess how these visual strategies impact teenage users, with consideration for psychological and cultural factors.

The methodology applied in this dissertation includes a qualitative, interpretive approach that combines visual analysis, semiotic theory, and design principle frameworks. Key platforms are examined through comparative interface analysis supported by academic literature and psychological studies. The research also incorporates cross-cultural insights into user perception of colour and visual structure.

The structural aspects of platform design equally contribute to shaping user experience. As outlined by Lidwell, Holden, and Butler (2003), principles such as consistency, alignment, and accessibility significantly enhance usability and user satisfaction. By encouraging a sense of mastery and comfort, structural clarity - the arrangement of menus, the hierarchy of information, and the predictability of navigation paths - not only reduces cognitive load but also raises emotional involvement. In digital environments saturated with competing stimuli, effective structural design serves as an anchor for user attention and trust. The importance of visual aesthetics in human-computer interaction has been highlighted by recent studies. According to Lavie and Tractinsky (2004), there are two main aspects of perceived online aesthetics: expressive aesthetics, which honour creativity and uniqueness, and classical aesthetics, which place an emphasis on order, balance, and clarity. Platforms like YouTube strategically balance these dimensions: a clean, familiar layout facilitates usability, while dynamic thumbnails, colourful branding, and customizable features inject expressive appeal. Platforms satisfy customers dual needs for convenience and emotional stimulation by combining both qualities.

Furthermore, research has shown that people frequently mistake utility for aesthetic appeal. Tractinsky, Katz, and Ikar (2000) found a robust correlation between the perceived beauty of an interface and its perceived usability, independent of actual functional performance. This "what is beautiful is usable" evaluation study highlights the potent influence of aesthetic design elements, particularly colour and structure, in shaping user expectations and behaviours. A visually appealing platform encourages longer engagement times, higher satisfaction, and greater loyalty, even when functional usability remains constant.

The dissertation also looks at how prominent video-sharing websites intentionally utilise colour and structural design to improve user experience. In order to show how visual components serve as crucial instruments of communication and control rather than just as ornamental accents, the study draws on semiotic theory, design principles, and psychology studies of perception. This dissertation is organised into four main chapters following this introduction. A historical and semiotic framework for understanding the cultural meanings of colour is established in Chapter One, which charts its development from religious and ceremonial symbols to modern digital situations. With a focus on emotional branding and neuromarketing, Chapter Two examines the intersections between branding tactics and colour theory in the context of interface design.

In Chapter Three, the main video-sharing platforms are compared, with an emphasis on how colour and structural decisions affect the user experience on sites like Vimeo, TikTok, and YouTube. Chapter Four explores how colour dynamics specifically affect adolescent user engagement while taking technological, cultural, and psychological aspects into account.

Through this investigation, the dissertation makes the case that colour and structure are essential to how platforms interact with, convince, and hold on to users rather than being merely decorative or incidental elements of platform design. Gaining an understanding of their strategic application may help designers create more efficient and morally conscious user experiences while also providing a better understanding of the power dynamics present in digital environments.

Chapter One: Colour, Culture, and Meaning: A Historical and Semiotic Framework

1.1 The Historical Role of Colour in Human Culture

Throughout history, colour has held profound significance far beyond its visual properties. It has served as an essential part of identity, religious symbolism, ritual, and social order. Before colour was used as a branding strategy or as a design feature in digital interfaces, it was a significant factor in how people understood the world around them. It was given significance by materiality, culture, and context. This chapter explores how colour has historically been used as a marker of spiritual belief, social class, emotional expression, and cultural identity, setting the foundation for understanding its semiotic and symbolic role in contemporary digital environments.

Many civilisations have traditionally used colour as a means of symbolic representation in their religious and ceremonial traditions. In ancient Greco-Roman rituals, specific colours were associated with the divine and used to distinguish different ritual roles. White garments, for example, were linked to purity and transition, while deeper colours such as red and purple marked sacred presence or royal status. Verity Platt (2021) notes that colour in these contexts was not decorative, but essential to conveying the distinction between human and divine realms and often functioned as a visual language embedded into religious dress, offerings, and sacred art.

The value of colour extended beyond symbolic meaning to material worth. The pigment ultramarine, which is derived from ground lapis lazuli, is among the most prominent illustrations of this. Highly sought after during the Middle Ages and Renaissance, ultramarine was costly due to its rarity and the difficulty of extracting the pigment. As Philip Ball writes, “Ultramarine is prepared from lapis lazuli... For centuries, all of Europe depended on the Afghan mines for their most precious pigment, which cost more than its weight in gold” (Ball, 2001, p. 76). Because of its extraordinary value, ultramarine was used sparingly, typically reserved for painting the robes of the Virgin Mary, where it served not only as a symbol of divine purity but also of the patron’s wealth and devotion, as seen in **Figure 1**.



Figure 1: The Virgin in Prayer by Sassoferrato, c. 1640. Use of ultramarine in Marian imagery signifying both sacredness and material value (National Gallery, 2024)

The meanings attached to colour have long been shaped by cultural context, often varying between societies depending on tradition, environment, and symbolic systems. Some connections seem universally recognised, like blue with calm or red with intensity, while others are different. For instance, white represents purity in most of the West yet is strongly associated with grief in many East Asian cultures. Similarly, red may represent festivity and good fortune in China, yet in Western media it may represent danger or caution. These differences show that colour is a culturally encoded system of meaning that is learnt, reinforced, and reinterpreted over time rather than a universal language.

Although this variety is evident in past usage of colour, current study shows that cultural differences in colour interpretation persist. People's understanding and reactions to fundamental colour categories varied significantly, according to a study by Jung et al. (2018) that looked at colour-emotion correlations in nine different nations. In certain places, red was linked to enthusiasm and vitality, but in other places, it was linked to danger or animosity. Similarly, perceptions of colours like green, yellow, and black shifted depending on regional

emotional codes. This evidence supports the idea that the meaning of colour is influenced by culturally learnt processes rather than being biologically set.

The communicative power of colour comes from its meaning's adaptability. Far from being instinctive, emotional responses to colour are shaped by repetition, narrative, and culture. Colour may serve as a visual shortcut to concepts, beliefs, or emotions when cultures gradually establish consistent expectations about what colours stand for. Patricia Sloane (1986) captures this dynamic precisely when she writes that colour symbolism often leads to “a predetermined response based on literary and psychological ideas about colour, rather than a response to the nature of colour itself”. The meaning of colour is in the stories we tell with it, not in the colour itself.

Recognising colour as a socially embedded system of meaning challenges the idea that it can be interpreted purely through preference or perception. Whether used in religious settings, artistic creations, or ceremonial attire, colour conveys values that are acquired, passed down, and interpreted contextually. These longstanding symbolic roles shape how colour continues to operate in contemporary visual culture.

1.2 The Semiotics of Colour

Colour, within a semiotic framework, is not merely visual decoration, but a structured and communicative sign system. It carries meaning in cultural, emotional, and functional dimensions, all of which are shaped by social convention rather than innate value. As Barthes (1968, p. 1) states, “all known cultures utilize signs to convey relatively simple messages swiftly and conveniently. Signs can have many meanings depending on their shape, position, colour, and environment. Colour, therefore, functions not in isolation but as part of broader systems of representation.

The construction of meaning in semiotics can be approached through two foundational models. The first, established by Ferdinand de Saussure, defines a sign as the union of a signifier (the sensory form) and a signified (the associated concept). Saussure emphasises that the link between these two elements is arbitrary, stating that “the bond between the signifier and the signified is arbitrary” (Saussure, 1983, p. 67). In this context, colour meanings—such as red connoting urgency or blue connoting calm—are not natural, but emerge through cultural and linguistic systems of difference and opposition.

The second model, developed by Charles Sanders Peirce, introduces a triadic understanding of signification comprising the representamen (the sign itself), the object (what the sign refers to), and the interpretant (the concept formed in the mind of the observer). According to Peirce, “anything can be a sign as long as someone interprets it as signifying something” (Chandler, 2007, p. 29). In this view, the meaning of colour is shaped not only by its form or use, but by the interpretive process through which it is understood, as illustrated in **Figure 2**.

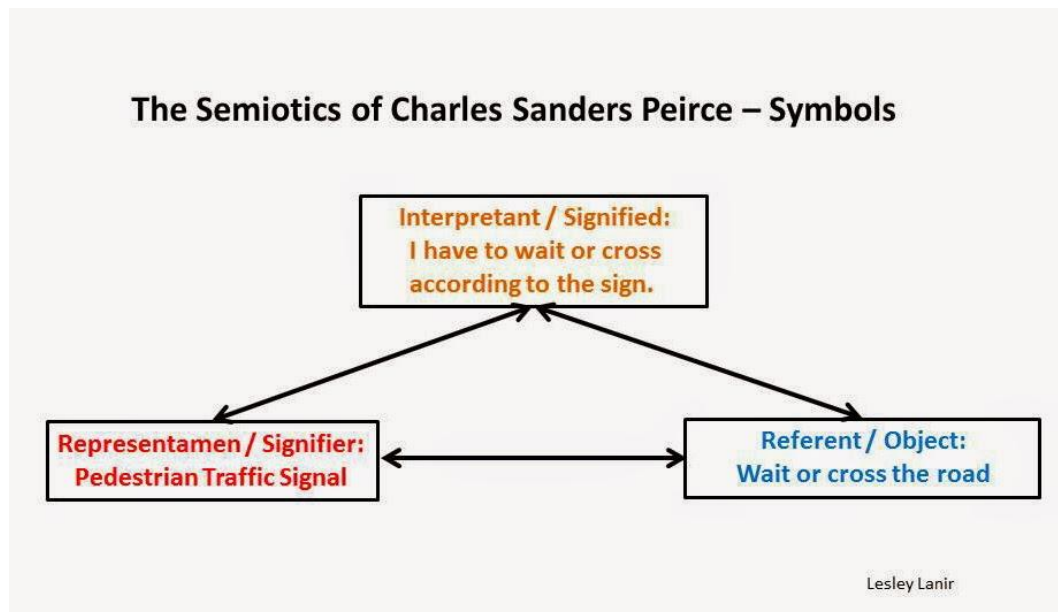


Figure 2: The Semiotics of Charles Sanders Peirce – Symbols (Lanir, 2019)

Peirce’s model also identifies three modes through which signs operate: iconically (by resemblance), indexically (through association), and symbolically (by cultural convention). Colour can function through all three modes depending on context and in most visual communication, it operates symbolically; however, indexical and iconic functions may also be present, particularly in cases where visual similarity or environmental cues inform interpretation.

Colour can be understood as a cultural and social mode—something that carries meaning not just visually, but within broader communicative structures. It functions within systems that assign value, identity, and emotion based on cultural context. This approach is reflected in the work of Kress and van Leeuwen (2002), who identify three metafunctions of colour: ideational, interpersonal, and textual. These allow colour to express ideas, influence the viewer’s position, and contribute to visual organisation. As they note, “the colours of flags, for

instance, denote specific nation states, and corporations increasingly use specific colours or colour schemes to denote their unique identities” (p. 347). In this way, colour becomes a strategic semiotic resource—integral to the construction of meaning across both media and social environments. In parallel, Eco (1985) describes colour as a “cultural unit,” highlighting that its meaning is not inherent but formed through shared codes and cultural conditioning (p. 159). A colour may appear neutral, but its interpretation is shaped by visual tradition, cultural memory, and communicative intent.

Colour, therefore, must be understood as an active component in the semiotic production of meaning. It operates within complex sign systems, shaped by history, perception, and cultural discourse. Its significance is never fixed, but continually negotiated through the interplay of context, convention, and audience interpretation. Recognising colour as a communicative agent—rather than a purely visual element—allows for a more critical analysis of how it is used strategically in digital spaces, particularly in branding and interface design.

1.3 Colour as Social Code and Identity Marker

Colour plays a crucial role in the construction of social meaning, operating not simply as a visual element but as a marker of group identity and belonging. Its function extends far beyond aesthetics, becoming a visible code through which individuals align with movements, traditions, or ideologies. Whether it is worn, shown, or institutionalised, colour influences how individuals react to others and represent who they are. It also serves as a visual indicator of affiliation in protest groups, converting philosophy into symbolic identification. The 2019 demonstrations by Extinction Rebellion illustrated this through the silent presence of the “Red Brigade,” whose members appeared in deep crimson robes and white-painted faces. They used colour as a symbol of environmental crisis and collective loss in their slow, ritualistic dances. As Lombardo (2021) notes, “the crimson red cloaks worn by these ghostly-white figures—the Red Brigade—symbolise the blood that all species on the planet have in common.” In this context, colour is performative as well as ornamental; it is meant to elicit strong feelings and attract media attention. This moment is captured in **Figure 3**, where the visual contrast of red against a grey city backdrop intensifies the message of sacrifice and collective vulnerability.



Figure 3: Extinction Rebellion's Red Brigade at a climate protest in London, using red as a symbol of collective grief and urgency (Reuters, 2019)

Historical movements like the suffragette movement similarly adopted a defined palette - purple, white, and green, not simply for branding, but to make political allegiance publicly recognisable. "The wearing of political colours is a significant statement of identity and/or values," as Sawyer (2007) contends (p. 146). These examples show how colour unifies participants while creating symbolic distance from the status quo.

Subcultural groups often use colour as a way of visually marking distance from dominant cultural values and creating a sense of internal cohesion. Black clothing is a key component of the Goth subculture's visual identity. As Fauquet-Alekhine (2016) notes, "Black color for clothes is one of the main binders that contributes to give consistency to this subculture," suggesting that colour functions as more than an aesthetic - it acts as a shared code that makes belonging legible. Style is how subcultures create their identities, and Hebdige (1979) examines how groups such as punks utilise "bricolage" to turn commonplace colours and apparel into symbols of resistance. The use of harsh black tones, safety pins, and tartan was transformed into a visual language of provocation and anti-establishment slogans. Colour selection in goth and punk transcends aesthetics and serves as a deliberate act of social difference and symbolic rebellion.

Gendered identity is often communicated visually before it is ever verbalised, and colour plays a central role in this process. Pink and blue are commonly utilised to differentiate between femininity and masculinity in Western cultures, especially throughout childhood. These assignments are reinforced by marketing, toys, and social media messages rather than being physiologically determined (Karmakar, 2023). These connections gradually influence perceived rules around conduct and identity in addition to aesthetic choice. Women tend to be more visually responsive to subtle colour variations, especially in green and blue hues—a difference that may inform both cultural representation and individual expression (Ellis & Ficek, 2001). Despite their seemingly innocuous appearance, these codings are part of a larger system where colour organises social expectations and emotional tone. This is especially noticeable in consumer culture, where visual language is still used to reinforce binary structures in packaging and regular product design. The recurrence of gendered colour schemes continues to influence how identity is seen, adopted, and socially interpreted, even in cases when the intentions are neutral.

In institutional settings, colour is used to standardise appearance, enforce roles, and shape public perception. Certain colour schemes are used in uniforms for occupations including law enforcement, aviation, and healthcare to convey authority, discipline, and trust. Professional uniform colour is not picked at random, but rather strategically in accordance with the ideals that each function aims to convey, as Lakshmi and Pooja (2018) describe. Dark blues and greys project reliability and seriousness in military and corporate settings, while white is commonly used in medical environments to suggest hygiene, transparency, and care. These selections are visual structures that educate the viewer and observer in addition to reflecting organisational branding. Incorporating colour into the institutional messaging aids in communicating hierarchy and establishing order. In this situation, it functions as a regulated signal incorporated into a larger framework of codes, authority, and expectations rather than as a means of personal expression.

Chapter Two: Colour in Branding and Platform Design

2.1 Colour in Branding, Emotion, and Consumer Recognition

In the emotional aspect of branding, where visual identity is created for connection as well as recognition, colour is crucial. Emotional branding aims to align a product with deeply personal and psychological needs. As Gobé (2001) explains, “emotional branding provides the means and the methodology for connecting products to the consumer in an emotionally profound way. It focuses on the most compelling aspects of the human character: the desire to transcend material satisfaction, and emotional fulfilment.” In this context, colour is not simply decorative - it is affective, sensorial, and symbolic. Whether it's a soft neutral palette suggesting wellness and care, or a vibrant, saturated tone signalling energy and youth, colour is deployed to resonate with emotions that shape how users perceive and respond to a brand.

The emotional power of colour in branding is not always consciously processed. Colour can affect decision-making before people are completely aware of it, according to neuromarketing study. This method looks for subconscious responses to visual stimuli using biometric technologies like eye tracking, face coding, and EEG. Mada (2024) highlights how colour alone can shift consumer choices in real-time, noting that even when individuals intend to behave one way, such as selecting a familiar product, they may change course in response to sensory cues embedded in the environment. This reactivity is enhanced in digital environments. A subtle shift in hue or contrast can alter a user's level of attention, trust, or emotional engagement. This implies that colour serves as a behavioural tool as well as an identity for businesses, influencing how customers engage with an interface or product on a sensory and cognitive level. Understanding the emotional effects of colour also requires a framework for identifying which emotions are most commonly triggered in consumer contexts. For this, Plutchik's Wheel of Emotions provides a helpful paradigm that assigns colour codes to eight fundamental emotions, including joy, fear, anger, and trust. The illustration sheds attention on how colour may either intensify or neutralise emotive reactions by highlighting the intensity and opposing pairing of feelings. This model supports the notion that colour serves as a psychological mechanism for emotional positioning as well as a branding strategy (**Figure 4**).

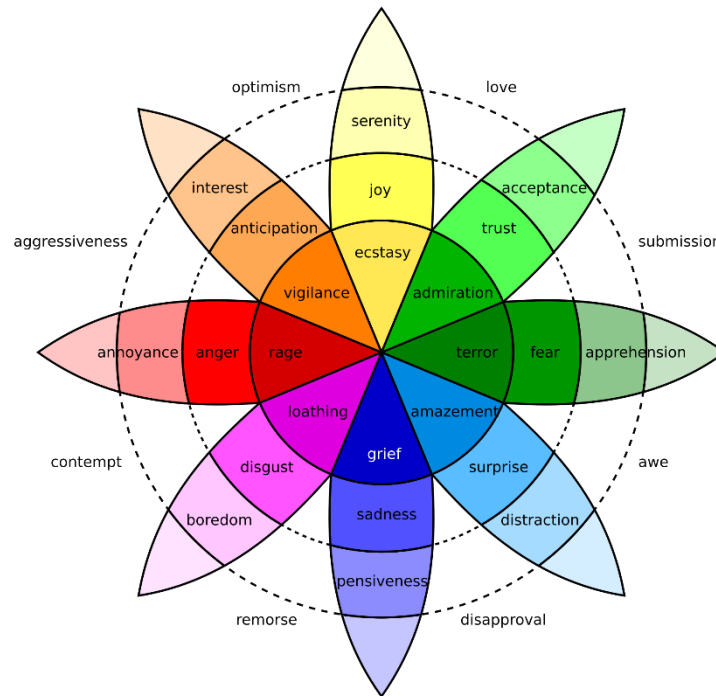


Figure 4: Robert Plutchik's Wheel of Emotions (Robert Plutchik, 1980)

Beyond emotional response, colour also plays a critical role in how brands are remembered. Colour serves as a visual anchor in highly saturated media situations, allowing for immediate identification before a name or brand is processed. Singh (2006) found that up to 90 percent of snap judgements about products are based on colour alone, highlighting its importance in brand differentiation and consumer recall. When used regularly, a brand's colour scheme acts as a sort of mental shortcut that builds trust and familiarity. Because people may come across a brand through a variety of touchpoints, such as app icons, thumbnails, websites, and advertising, this impact is particularly significant in digital environments. One example of how colour helps with memory and consistency in an overstimulated digital environment is the ability to recognise a platform by its main colour, such as red for Netflix or green for Spotify.

2.2 Platform Interfaces and the Logic of Visual Hierarchy

The interface design of digital platforms relies heavily on the strategic use of colour and structural organisation to guide user attention, enhance usability, and communicate functionality. Beyond its cultural and emotional meanings, colour is essential for identifying interactive features, communicating navigation signals, and sustaining user interest in visually complex settings.

In order to improve digital interfaces' clarity and usefulness, colour contrast must be used effectively. According to Aloumi (2013), users may more easily recognise actionable regions and key content when there is a suitable contrast between background and foreground components. In environments saturated with multimedia elements, the ability of colour to separate important items from decorative or secondary information becomes crucial. Colour coding functions as an understandable visual language that users decipher almost instantly. For example, red indicates urgent messages, whereas grey indicates disabled options.

Alongside colour, visual hierarchical structure has a big impact on how people digest and navigate information. Gestalt theory offers a fundamental structure for comprehending these design ideas. As Smith-Gratto and Fisher (1999) argue, principles such as figure-ground distinction, proximity, similarity, and closure govern how visual elements are perceived and grouped. While proximity and similarity direct the user's eye towards related elements, facilitating more effective information retrieval and interaction, clear figure-ground connections guarantee that text and symbols stand out against their backgrounds.

Additionally, visual hierarchy is consistent with typical eye-scanning tendencies seen in users of online interfaces. Users frequently interact with material in accordance with identifiable scanning patterns, particularly the F-pattern and Z-pattern (**Figure 5**), according to the Interaction Design Foundation (2023). The F-pattern is commonly seen on pages with a lot of material, such news websites, where people read across the top horizontally at first, then scan down a little then read across again before scrolling vertically down the left. By directing the eye over the top, diagonally down to the other side, and finally horizontally across the bottom, the Z-pattern, on the other hand, is frequently found on simpler interfaces. The strategic arrangement of important visual components to optimise visibility and user engagement is informed by these natural eye movement patterns, which are crucial factors in interface design.

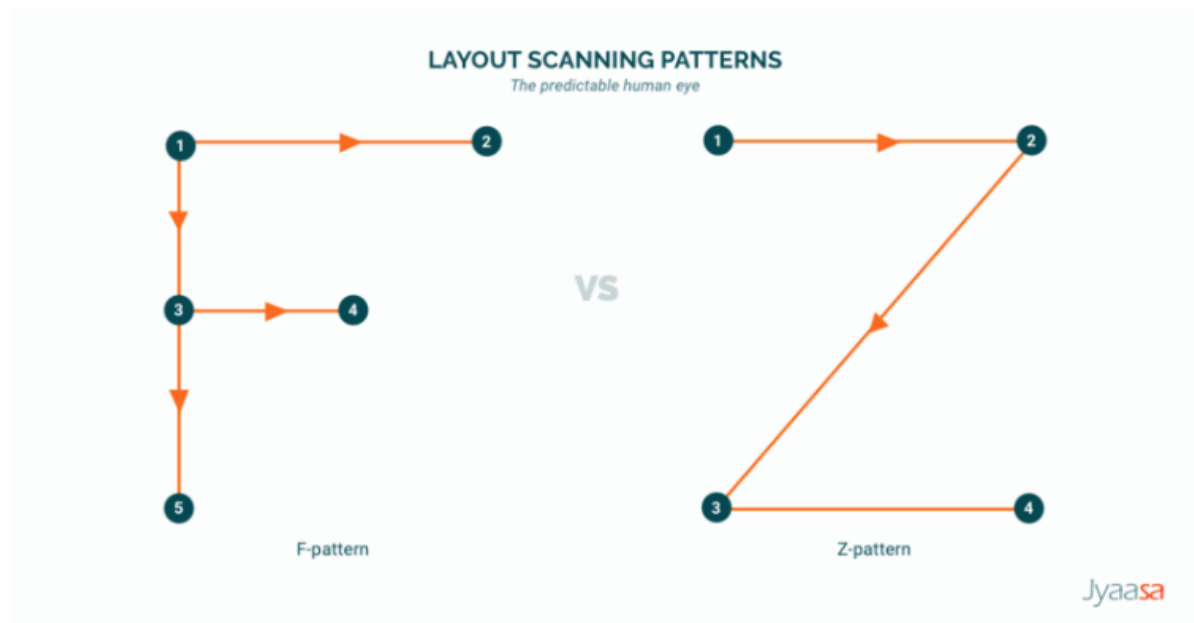


Figure 5: Diagram comparing F-pattern and Z-pattern scanning behaviours (Medium, 2019)

Furthermore, platform interfaces that align with established usability heuristics tend to foster greater user satisfaction and engagement. Jakob Nielsen's principles, particularly those concerning visibility of system status, consistency, and minimalist design, reinforce the importance of clear and intuitive visual hierarchies (Nielsen, 2006). Minimalist layouts that prioritize essential information reduce cognitive load and enable users to navigate interfaces more fluidly, particularly when colour and spatial arrangement are used consistently to signal affordances and actions. Overall, the interaction between colour application, visual hierarchy, and natural eye movement patterns creates the navigational and cognitive scaffolding that users rely upon when interacting with digital platforms. These guidelines not only improve visual appeal but also organise the user experience to encourage predictability, clarity, and usability.

Chapter Three:

Platform Comparison - Visual Branding and Colour Use in YouTube vs Competitors

3.1 Visual Branding and Colour Cues in Video-Sharing Platforms

User perception and engagement are significantly influenced by the branding and interface design decisions made by digital video-sharing platforms, especially in environments that are becoming more visually and algorithmically mediated. Among the several visual elements that make up a platform's identity, colour serves as a semiotic mechanism that conveys mood, purpose, and behavioural direction in addition to being an aesthetic technique. It organises the user's emotional and navigational experience while capturing brand identity.

Despite frequent comparisons between platforms like YouTube and Vimeo in terms of user base, accessibility, and content culture, their approaches to visual communication, particularly regarding colour, display different strategic goals. As Lee and Kim (2014) observe, "the current user interface design of video-sharing sites is not simply the interface design of the video player, but rather the system of the site itself, encompassing colour, layout, navigation, and behavioural feedback" (p. 2). This suggests that colour choice is an essential part of the platform's communication architecture rather than a stand-alone design feature.

YouTube's reliance on a red-based palette operates as a purposeful psychological trigger, evoking urgency and encouraging rapid interaction, particularly through visual signals such as the subscribe button and progress bar. Red, which is frequently linked to excitement, alertness, and decisiveness, supports YouTube's positioning as a dynamic, fast-paced site where user experience is mostly dependent on immediacy and attention retention. Vimeo, on the other hand, has a more restrained and professional feel, centred on cold colours like blue and grey, which complements its positioning as a platform for creative professionals looking for an interface devoid of distractions. As Ricard (2015) notes, the tone transmitted by each platform's visual environment is not only a design decision, but a communicative statement of purpose: YouTube conveys accessibility and immediacy, while Vimeo cultivates a sense of focus, exclusivity, and elevated creative intent. These identification signals influence user behaviour and perception, therefore they are not neutral.

When comparing the interface designs of each platform, this visual discrepancy becomes clear. While Vimeo's focus on negative space, distinct visual hierarchy, and minimum distractions enable a more controlled, immersive experience, YouTube's bright red accent and crowded sidebar stimulate engagement and quick navigation (**Figure 6**).

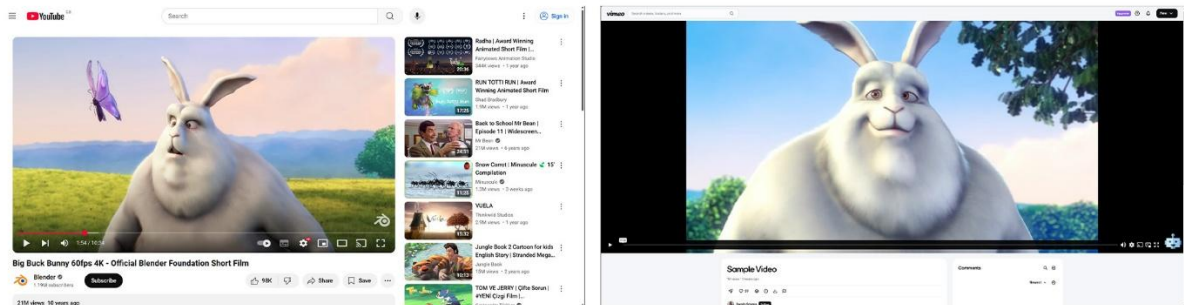


Figure 6: Comparison of YouTube and Vimeo's video player interfaces, highlighting differences in colour use and visual hierarchy (YouTube & Vimeo, 2025)

In both cases, colour acts as a visual signifier—encoding platform values, user expectations, and interactional tempo. It operates within a larger visual grammar that not only differentiates one platform from another but also directs the nature of engagement, emotional tone, and the overall user journey

3.2 Colour as a Tool of Interface Navigation: Comparative User Interaction Design

In video-sharing platforms, colour is not only an aesthetic tool but also a functional one - guiding users, signalling engagement opportunities, and shaping the rhythm of interaction. Colour is used in distinctive ways on platforms like YouTube, Vimeo, TikTok, and Instagram to promote habits and maintain focus. This section compares the design logics of each platform and examines how colour functions inside the user interface.

The interface of YouTube is bold and highly contrasted. Its signature colour is red, which can be found in the progress bars, notification icons, and "Subscribe" button. These colour triggers are not random—they are designed to stand out against a clean background and prompt immediate action. However, this effectiveness can be hindered by the platform's visual density. The clarity of colour signals may be limited by the large number of thumbnails, advertisements, comments, and suggestions on YouTube's interface. As Cabrera et al. (2021) note, user engagement on YouTube is increasingly shaped by visual cues and algorithmic feedback, and the platform's layered structure has led to the development of new metrics - such as conversation, applause, and amplification, used to

track interactivity beyond simple views (p. 5). This supports the notion that colour is a component of a more intricate visual system.

Burgess and Green (2009) also describe YouTube as a hybrid platform where professional and amateur content, commercial interests, and community engagement coexist: “YouTube is experienced in a range of different ways by consumer-citizens via a hybrid model of engagement with popular culture – part amateur production, part creative consumption” (p. 14). This hybridity is supported by the colour-coded interface, which enables users to discern between activities (such as commenting, like, and subscribing) while accessing both interactive and entertaining material.

Vimeo, by contrast, offers a more restrained experience. Cool colours like blue and grey are used throughout its interface to create a serene, understated feel. Where YouTube’s interface is built for constant decision-making and quick clicks, Vimeo encourages slower, more focused engagement. The use of colour is subdued yet effective; harsh suggestions are avoided, and buttons are highlighted only when needed. This is consistent with Vimeo’s standing as a platform for professionals, artists, and makers. Users are encouraged to explore material more purposefully and without interruption thanks to its appeal.

TikTok adopts a completely different approach. The app’s black background, paired with vibrant icons—such as red hearts, electric blue comments, and swirling effect buttons—creates a hyper-stimulating environment. On TikTok, colour, motion, and music combine to produce a cycle of micro-engagements. Pourazad, Stocchi, and Narsey (2023) found that while platforms like YouTube and Facebook show a “reverse double jeopardy” effect—where influencers with larger followings often receive lower engagement—TikTok tends to reward immediate interactivity regardless of size, with higher visibility for content creators using emotionally and visually dynamic strategies (p. 140). One of the main forces behind these micro-engagements is colour, which is employed to highlight interaction cues and increase emotional immediacy.

Instagram Reels combines the immediacy of TikTok with the simplicity of its parent platform. Brighter colour contrasts, such as animated hearts, comment boxes, and story rings, are included into the Reels interface, which maintains a clean overall look while visually differentiating short-form video from other content. Depending on the format, colour can be utilised contextually to convey urgency, emotional tone, or engagement. The signals are nevertheless crucial for sustaining a quick visual rhythm across user journeys, even if they are less popular than on TikTok.

What unites these platforms is not the specific palette, but the strategic use of colour to communicate function. YouTube's red stands out in a busy interface; Vimeo's understated blues focus attention; TikTok's saturation evokes strong emotions; and Instagram's colour palette responds to context. Each instance uses colour as an interface language - a grammar of engagement that impacts user behaviour, feeling, and connection with the platform.

3.3 Emotional Retention and UI Strategy in Video Platform Design

Emotional retention is now a central goal in the interface design of video-sharing platforms, especially as content becomes shorter, more dynamic, and increasingly competitive. Small design decisions, such as icon placement and animation speed, as well as layout simplicity, help organize not only how users interact but also how they feel and, eventually, how long they stay. Although the interfaces for short-form videos on TikTok and YouTube Shorts may appear to be comparable, they differ greatly in terms of emotional pace, engagement logic, and interaction dynamics, all of which are indicative of deeper strategic objectives in user retention.

Li Hao and Kim (2023) note that both TikTok and YouTube Shorts use a “vertical ‘3×N’ thumbnail arrangement,” where three thumbnails are displayed per row on the user’s profile page (p. 2). As shown in **Figure 7**, while the layout is structurally similar, the way social metrics are presented creates a different emotional atmosphere. On YouTube Shorts, only the view count appears within the thumbnails, while subscriber count and the total number of videos uploaded are displayed at the top of the page. This keeps the interface relatively minimal and focused. TikTok, by contrast, places more emphasis on creator status by showing the total number of likes received, along with follower and following counts, directly above the video grid. These metrics frame the content with visible performance indicators. As Li Hao and Kim further explain, this interface approach “incites competition between creators,” encouraging users to compare popularity and engagement at a glance (p. 2)

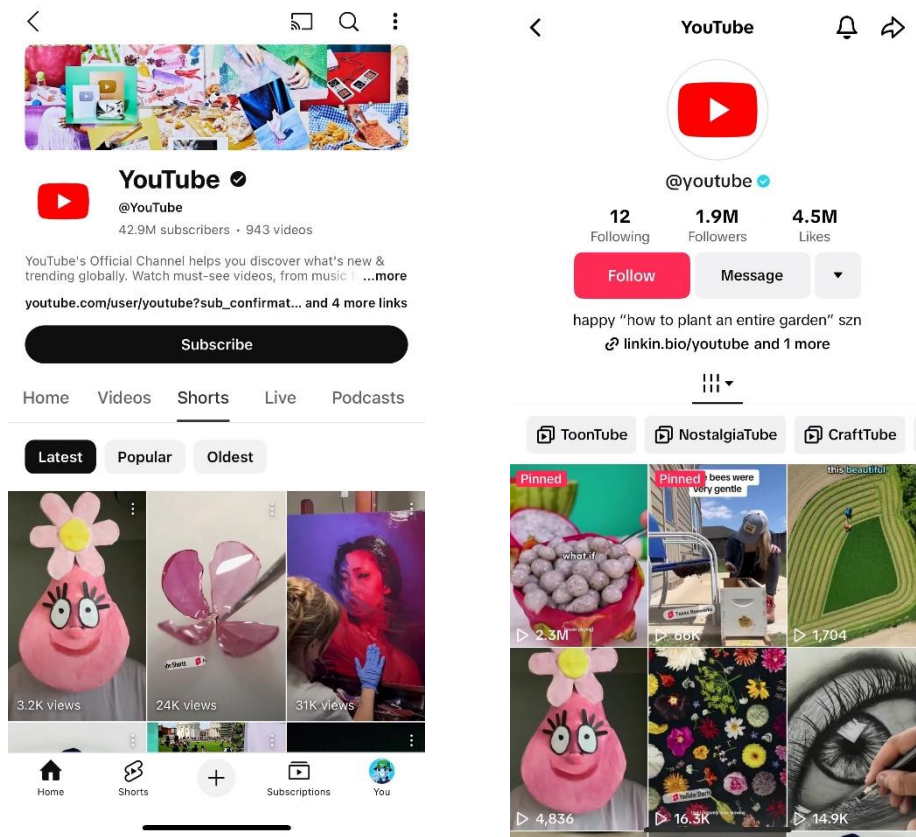


Figure 7: Side-by-side comparison of TikTok and YouTube Shorts profile layouts, highlighting differences in thumbnail arrangement and social metric displays. (YouTube & TikTok, 2025)

This competitive energy becomes even more apparent within TikTok's video feed. Fast-paced, full-screen content is the foundation of its interface, which uses swiping and constant feedback to keep users engaged. A new video appears with every gesture, surrounded by animated icons and engagement prompts. Zannettou et al. (2024) found that over time, users on TikTok became more interactive—liking more content and spending longer on the platform - yet their attention to individual videos remained low. “Most participants watch between 30% and 50% of all videos till the end,” the authors report, noting that the platform “prioritizes increasing the time spent on the platform and user engagement (through liking) rather than making recommendations that are likely to be watched until the end” (p. 10). Emotional retention, in this case, is maintained less through focused viewing and more through habit, motion, and reward.

YouTube Shorts mirrors this vertical format but introduces it within a calmer interface. The design language is still in line with the YouTube experience as a whole, feedback icons are smaller, and transitions are less abrupt. This creates a feeling of brand familiarity and visual uniformity on the platform. Rather than demanding attention through urgency, YouTube

encourages retention through ease of use and emotional predictability - users know what to expect, and that comfort supports longer-term engagement.

Among these, Vimeo offers a striking difference. Its UI makes a case for emotionally restrained design, while not being a short-form platform. Performance metrics are hidden, discovery is shaped by editorial curation, and the visual environment is clean and intentionally subdued. Vimeo is "a community of video lovers... dominated by the thirst of discovery and learning, rather than fame," according to Bietti (2012) (p. 9). This method emphasizes how clarity and focus, as opposed to continuous feedback, may maintain emotional involvement.

Comfort-oriented features like Dark Mode also contribute to emotional retention by reducing visual fatigue and supporting long-term usability. Erickson et al. (2020), in a study on immersive virtual environments, found that dark-themed interfaces significantly lowered eye strain compared to light modes—especially in low-light settings. Participants also demonstrated higher visual acuity and expressed a preference for dark mode when reading text under dim conditions. While their research focused on VR headsets, the findings align with how platforms like YouTube implement theme flexibility, offering users control over the visual environment and reinforcing emotional comfort through adaptable design.

When combined, these platforms show a variety of emotive tactics. The layout of YouTube places it in between the serene, carefully regulated experience of Vimeo and the intense feedback loop of TikTok. Its interface supports emotional retention by familiarity, restraint, and visual comfort - not through overstimulation, but through usability and trust.

Chapter Four: Colour Dynamics in Teenager Engagement

4.1 A Brief Introduction Applied to Colour Psychology

Colour is more than just a decorative element used in visual communication; it is a powerful psychological trigger that influences behaviour, emotion, and perception. Colour is purposefully utilised on digital sites such as YouTube to direct attention, create ambiance, and affect user engagement. Because of their increased emotional sensitivity and developmental stages, adolescent users might be more affected by colour. Teenagers, as digital natives, are continuously exposed to visual stimuli, making them particularly sensitive to emotional clues inherent in interface design. Recent research also highlights how higher colour complexity, the use of varied and richly distributed colours in digital images—can serve "as an external cue that draws social media users' attention and provokes a shift from the users' peripheral processing mode to a central processing mode" (Kanuri, Hughes & Hodges, 2023, p. 2), increasing engagement, especially in fast-paced social media environments.

Colour psychology, which investigates the emotional and cognitive effects of colour, is frequently used in areas such as branding, advertising, and user interface (UI) design. According to Elliot and Maier (2014), various colours can elicit consistent psychological responses - for instance, the colour red can raise arousal and attentional concentration, whereas blue might promote peace and trust. These impacts are amplified in digital situations, where colours serve as both aesthetic and behavioural signals. Bonnardel, Piolat, and Le Bigot (2011) found that colour had a considerable influence on users' decisions of website aesthetics and browsing behaviour. Warmer tones enhanced time spent engaged with the information, but cooler tones improved work efficiency and lowered cognitive burden. Their findings reveal that colour choices may have a significant impact on both psychological reactions and functional outcomes – an especially important consideration when designing for teenagers, whose emotional responses and browsing habits are heavily influenced by visual design.

4.2 Emotional & Behavioural Responses

Teenagers' emotional sensitivity and ongoing cognitive development make them particularly responsive to the visual elements of digital platforms. Colour has a significant impact on how people engage with and understand online content. This impact derives not just from psychological connections, but also from the fact that teens are still developing emotional attachments to visual stimuli. Their colour choices and emotional responses are frequently flexible, influenced by peer culture, trends, and personal identity development in digital

contexts. Rather of responding to colour with a set emotional intensity, teenagers link colour with a greater spectrum of emotions but with less intensity compared to older age groups. Jonauskaite et al. (2024) discovered that individuals aged 16-19 had more diversified emotional responses to different colours, although they were less consistent or strongly felt. This shows that emotional meaning is still developing during adolescence, which might explain why visual design has such a large behavioural effect in this age range. As platforms like YouTube rely extensively on colour to indicate urgency, mood, or tone - in thumbnails, progress bars, or user interface components - this age group's reaction becomes more important.

The reasons for these colour choices are both developmental and environmental. According to the Ecological Valence Theory, individuals tend to favour colours that correspond to positive experiences throughout their lives (Palmer & Schloss, 2011). Colour choices for teens, who are still developing emotional associations, may be more varied and significantly influenced by current digital environments and social interactions. A colour palette utilised consistently throughout emotionally satisfying content, such as thumbnails from a particular creator or a trend style, might start to form emotional resonance and promote engagement behaviour.

Evidence from platform-based studies supports the psychological significance of colour in teen interactions. Yousefi, Cakmak, and Agarwal's (2024) multimodal examination of user responses to YouTube material discovered that visual colour features had a greater emotional impact than auditory or written components in specific video genres, notably movie trailers. Warm tones, such as red and orange, were more commonly connected with high-arousal emotions like excitement and stress, whereas cooler hues were associated with peace or melancholy. These findings imply that colour has a major impact on emotional expectancies, particularly in contexts where material is processed fast, and attention is constrained.

This responsiveness to colour among teenagers highlights the need for intentional visual design when targeting younger audiences. Their emotional interpretations are not permanent, but rather moulded by repeated exposure and social environment, particularly on platforms where colour plays an important role in material display. Colours used to convey mood, urgency, or identity inside interfaces become more than just a design option; they also have a behavioural effect that demands greater examination when examining specific platforms like YouTube. resonant and developmentally acceptable for adolescent users.

4.3 Application of Colour on YouTube's Interface

On YouTube, colour serves a vital role in shaping user experience, directing attention, and influencing emotional response. Colour becomes an essential tool for directing interaction and boosting engagement, especially for adolescent users whose media consumption patterns are frequently fast paced, visually led, and emotionally responsive. This is evident in YouTube's user interface, visual design language, and content formats, all of which are made to be emotionally compelling, intuitive, and convincing.

In 2023, YouTube's design team unveiled a redesigned version of the platform's signature red, which has long been an identifiable component of its visual identity. The new version is somewhat softer and more sophisticated, with a gradient that leans towards magenta. According to Google Design (2023), when designing the new gradient, the team “considered several colours adjacent to red” and ultimately selected magenta “for its connotations of imagination and evolution.” **Figure 8** visually demonstrates this transition across the YouTube colour spectrum.

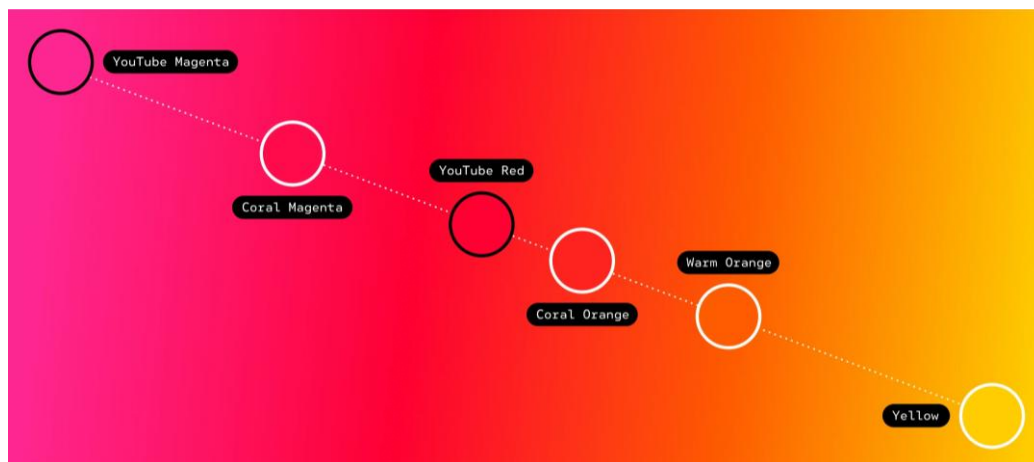


Figure 8: YouTube's new gradient, transitioning from red to magenta (Google Design, 2023).

The new tone was developed to seem more lively, approachable, and emotionally balanced across all devices and lighting conditions. Instead of showing up everywhere, the colour is used selectively in strategic places like the progress bar, subscribe button, and notification alert. This keeps the colour's psychological effect intact by serving as a signal to take action or interact. These red signs assist to direct attention and promote platform habits like clicking, subscribing, and stopping for younger users by standing out in the otherwise neutral UI.

Thumbnails, one of the most powerful visual elements on the website, also depend heavily on colour. According to Seungmin Lee (2023), thumbnails significantly affect user happiness and video selection, especially when they include strong colour schemes, emotive images, and a distinct typographic hierarchy. Users may choose what to watch even before reading the title or description thanks to these visual signals that grab their attention and provide an emotional hook. These thumbnails serve as quick, emotionally charged visual invites for adolescent users, whose online conduct is frequently emotionally motivated and impacted by peer culture and aesthetics. Lee further underlined the importance of visual language as a shortcut for interpretation by pointing out that colour not only affects perceived quality but also aids users in rapidly determining the tone or intensity of a movie. Supporting this, Liu (2023) also highlights the critical importance of thumbnail design, arguing that effective thumbnails directly influence viewer engagement and channel growth (see **Figure 9** for an example of a high-performing thumbnail following these principles).

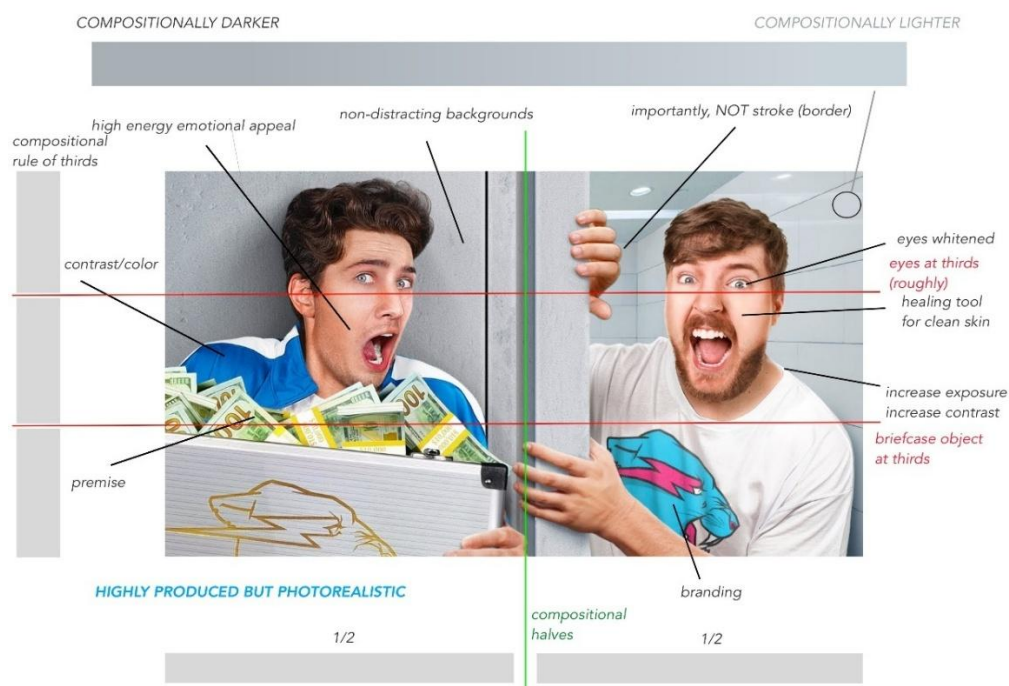


Figure 9: High-performing YouTube thumbnail design (Liu, 2023)

Visual responsiveness is now even more important thanks to recent platform format changes, especially the launch of YouTube Shorts. Vibrant colours, quick cuts, and emotive imagery are key components of shorts, which are made for vertical, mobile-first viewing and keep viewers interested. According to Violot et al. (2024), short videos receive a lot more views and likes than normal uploads; this impact is particularly noticeable in categories that

are orientated towards entertainment purposes. Their research showed that the Shorts format's visual simplicity, together with its swipe-up interface and vibrant colour palettes, created a high-engagement environment that is ideal for brief attention spans and rapid emotional reactions. Even while the study found that Shorts generate fewer comments per watch, indicating more passive involvement, their popularity is still largely due to their dependence on vivid, emotionally powerful images.

YouTube visual tactics are also related to more general engagement strategies. Ego-focused design methods, or those that appeal to one's self-image or personal identity, have been shown to increase engagement on YouTube, especially in tech-related advertisements (Wang, Yang, & Haught, 2023). The study demonstrated how visual components support emotional resonance and brand trust, while without having an explicit focus on colour. These results lend credence to the notion that colour, as a component of the larger visual language, improves behavioural activity as well as psychological responses.

4.4 Cultural and Contextual Considerations in Colour Design

While colour has universal psychological effects, its meaning and emotional impact can vary significantly across cultural and contextual boundaries. Colour must serve as both a visual attraction and a culturally flexible tool for an international platform like YouTube, which serves a broad adolescent audience. Something that feels bold and exciting in one place could be viewed as inappropriate, neutral, or even negative in another. Assessing YouTube's design decisions and how they affect user perception and interaction across cultural boundaries requires an understanding of these subtleties.

Colour meanings are not fixed — they are shaped by shared cultural histories, religions, ideologies, and local aesthetics. For instance, in many Western cultures, red may indicate danger or urgency, while in Chinese culture, it is associated with good fortune, joy, and celebration. In a comparable way, white can symbolise purity in Western cultures, yet in some East Asian cultures, it is frequently connected to loss and grief. These connections are not arbitrary; rather, they result from social and symbolic structures that are deeply rooted (Kommonen, 2014).

Cross-cultural study indicates considerable constancy in colour choices despite these variances, especially when it comes to utility and visual appeal. Madden, Hewett, and Roth (2000) conducted a comprehensive multi-country research and discovered that while some colours, like blue, are generally well-liked worldwide, others vary according on the product

category and setting. This helps to explain why websites like YouTube tend to have simple interfaces with neutral hues like white, black, and grey and selectively incorporate accent colours like red to keep a worldwide design language without offending local viewers. However, colour on YouTube is not only experienced through interface elements; it is also deeply embedded in thumbnails and video aesthetics, which are often shaped by regional creator styles. Local preferences may be reflected in content from other areas. For example, Western entertainment thumbnails tend towards dramatic contrasts, close-up facial expressions, and saturated tones, whereas Korean or Japanese thumbnails typically contain brighter pastels and layered typography. These variations suggest that viewer expectations and regional design trends have a greater impact than algorithmic standardisation alone.

This contrast is clearly illustrated in Figure 10, which compares thumbnails from KonMari's Western and Japanese YouTube channels. The Western thumbnails are minimalistic, using neutral backgrounds, limited text, and calm visuals, while the Japanese thumbnails adopt a much more colourful, text-heavy, and layered visual approach. As discussed by Unseen Japan (2020), this difference reflects broader marketing strategies, where Japanese audiences tend to favour vibrant, information-rich visuals that feel approachable and energetic. The use of bold text overlays and playful design in the Japanese thumbnails aligns with common tropes found in Japanese television and advertising, reinforcing familiarity and visual engagement.



Figure 10: Comparison between KonMari's Western and Japanese YouTube thumbnails (Unseen Japan, 2020).

Colour interpretation is also influenced by context. Whether the video is political, entertaining or educational, its emotional tone can influence how colour is utilised and interpreted. High saturation and emotional images may work well for amusement material, while muted colour schemes are frequently used to convey tone and meaning in more sombre or serious subjects. According to semiotic theory, colours can function either symbolically or indexically, indicating deeper meanings beyond just aesthetics, as Darrodi (2012) argues. In this approach, designers may use colour not only to attract attention, but also to communicate cultural or emotional meaning across different types of content. Although YouTube applies a mostly universal interface design, creators often localise their visual language to suit their cultural context. Colour may serve as both a local and global language thanks to its hybridisation, which combines platform consistency with regional adaption. It represents how people perceive, understand, and react emotionally to images depending on their background and identity in addition to design principles (Garett et al., 2016).

Conclusion

To conclude this study, the research set out to investigate how the strategic use of colour and structure within the interfaces of dominant video-sharing platforms, particularly YouTube, impacts user perception, emotional engagement, and behavioural patterns. As outlined in the introduction, design was approached not merely as an aesthetic exercise, but as an act of communication shaped by deep user understanding (Norman, 2013). Through the course of this dissertation, it became evident that colour and structural layout play an integral role in guiding attention, eliciting emotional responses, and shaping patterns of navigation and retention.

The research has shown that colour and structure function not only as visual enhancements but as powerful semiotic devices that influence how users interpret, interact with, and remain engaged within digital environments. Platforms such as YouTube strategically use bold colour schemes and hierarchical structures to create intuitive yet emotionally charged experiences. Building on Barthes' (1968) concept that signs carry cultural meaning rather than natural value, this study demonstrated that the visual elements embedded within platform design communicate specific emotional cues and behavioural expectations to users, often below the level of conscious awareness. A key part of the study involved comparing the visual and structural strategies across different platforms, including YouTube, TikTok, Vimeo, and Instagram. This comparative study demonstrated how each platform employs certain colour schemes, layouts, and interaction patterns to create unique emotional and behavioural contexts. While TikTok uses hyper-stimulating, high-saturation designs to enhance short-term engagement, sites like Vimeo employ a more muted, professional look, and YouTube promotes urgency and familiarity through bright, high-contrast images. These variations show that colour and structure are carefully applied to platform identity, audience expectations, and emotional results rather than being applied generically.

The results also highlighted the importance of emotional retention techniques, especially in settings like TikTok and YouTube Shorts where users' attention is maintained through quick cycles through the use of colour, motion, and visual simplicity. Burgess and Green's (2009) characterisation of YouTube as a hybrid space, balancing amateur creativity with commercial interests, proved particularly relevant, underscoring how interface design simultaneously promotes participation and consumption through visual and structural cues. Moreover, the research explored how different user demographics respond to these strategies, with particular attention given to adolescent audiences. As Jonauskaitė et al. (2024) observed, teenagers display heightened but less consistent emotional responses to colour stimuli, making them especially receptive to the emotional framing embedded in

thumbnail imagery, layout organisation, and interactive elements. These results highlight how crucial it is to acknowledge visual design's developmental and psychological effects on various audience segments in addition to its practical impact.

While this study provides valuable insights, it also recognises its limitations. The qualitative, interpretive methodology inevitably carries a degree of subjectivity, and the rapidly evolving nature of digital interfaces presents challenges for long-term generalisability. Nonetheless, by synthesising semiotic theory, emotional branding strategies, and psychological insights within a comparative framework, this research offers a meaningful contribution to understanding how visual environments are crafted to influence digital behaviour.

In summary, this dissertation has demonstrated that colour and structure are not ancillary aspects of digital design but foundational elements that shape perception, emotion, and interaction within video-sharing platforms. Understanding their strategic deployment, both within individual platforms and across the broader digital ecosystem, not only deepens awareness of how digital experiences are constructed but also raises important ethical considerations for future design practice in an increasingly emotionally engineered media landscape.

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